Report Title

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1 Introduction

1.1 Ordered External Electric Fields as catalysts

Chemical reactivity as a field of study can be broadly reduced down to the dynamics of electrons, it therefore goes without saying that being able to manipulate electrostatics to control reactivity is a grand challenge of chemistry. This notion has been at the forefront of chemical research for as long as electron directing groups have been utilised to manipulate the strength of bonds, however the idea of influencing a reaction from a distance using electrostatics has only been explored in the past century.

The methods to this approach were initially quite crude, involving ionic aggregates in solution to polarise reactant species,[1] though through the use of early *ab initio* methods were able to be more specifically investigated. These later studies in the early 90s sought to understand solvent effects and more generally the effect of, localised electric fields,[2] and as a result, a more concrete understanding of the mechanism of bond polarisation was identified,[3] forming the foundations of the study of electric field catalysis.

$$\Psi_{A-B} = C_1(\mathrm{A{-}B}) + C_2(\mathrm{A^- B^+}) + C_3(\mathrm{A^+B^-}) \tag{1}$$

Following the work of Sini et al.[3] who described the electron pair bond as the superposition of their spin paired covalent and two possible zwitterionic forms (as shown in Equation (1)), and many subsequent studies that ill be outlined later in this report, Shaik et al. formalised the "governing effects and rules for electric-field effects on chemical reacivity".[4] Briefly summarised, the rules are as follows:

(a) An ordered external electric field (OEEF) that is oriented along a "bond axis" will polarise the bond, favouring one of the ionic species (Equation (1)), eventually cleaving it heterolytically. As this polarisation occurs, the bond dipole increases, causing it to interact more strongly with the OEEF, stabilising the species in the process.

(b) Every reaction possesses a "reaction axis" which follows the flow of electrons from reactant to product, through the transition state (TS).

(c) Applying an OEEF along the "reaction axis" will stabilise the TS by increasing the bond ionicity (as per rule Shaik (a)) and, (symmetry allowing) finish this off

(d) molecule will align with OEEF

##

– harnessing the power of OEEFs for catalysis

- introduce polarisation

- Bonds as linear combination of resonance contributors - Sini et al.[3] not sure if this is too fundamental to start with

$$[\mathrm{AB} \longleftrightarrow \mathrm{A}^{+}\,\mathrm{B}^{-} \longleftrightarrow \mathrm{A}^{-}\,\mathrm{B}^{+}] \tag{2}$$

$$\Psi_{A-B} = C_1(\mathrm{A{-}B}) + C_2(\mathrm{A}^{-}\,\mathrm{B}^{+}) + C_3(\mathrm{A}^{+}\mathrm{B}^{-}) \tag{3}$$

be aresounding yes!

- Shaik's governing rules - Shaik et al.[4]

- EEF catalysis for tethered molecules (STM) - Aragonès et al.[5]

- IEFs from enzymology - Warshel et al.[6]

1.2 Scaling OEEFs to Bulk Catalysis

- Marcus theory and reducing reorganisation energy - Landau[7]

– Breaking the solvation shell - Novák et al.[8]

- Restructuring of the solvent with an EEF - Daniels et al.[9]

- EEF "tweezing" effect/aligning of reacting species (Shaik rule 4) - Wang et al.,[10] Shaik et al.[4]

$$\Delta E = 4.8\vec{F} \cdot \vec{\mu}\,\mathrm{kcal{\cdot}mol^{-1}} \tag{4}$$

- Screening of the EEF by the solvent - Xu et al.,[11] Shida et al.[12]

- Using pulsed EEFs to reduce electrochemical reactivity - Xu et al.,[11] Shaik et al.[4]

- Specialist equipment required to mitigate polar solvent screening - Xu et al.,[11] Shaik et al.,[4] Shida et al.[12]

1.3 Creating IEFs from Ordered Ionic Liquids

- ILs - MacFarlane and Seddon[13]

- PILs - Greaves and Drummond[14]

- polarisability (leads to the IEF) need ref - Brehm et al.[15] (delocalisation of e-density)

- ILs slow migration makes them good candidates for long lasting IEF bearers with pulsed EEFs need ref

- ILs ordering in EEF - Zhao et al., Wang[16,17]

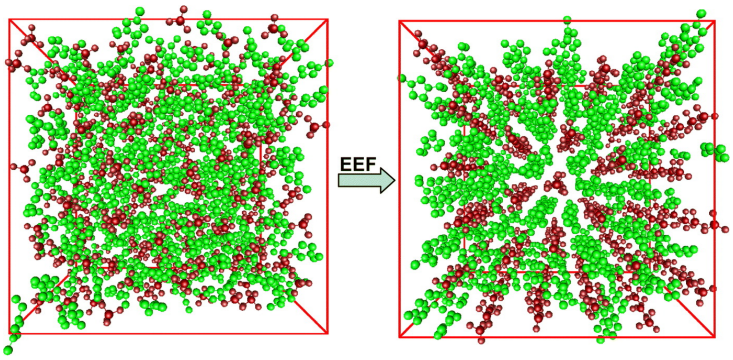

Figure 1.1: Snapshot from a molecular dynamics simulation of [C_2mim][BF_4] without (left) and with an external electric field of $1.14\,\mathrm{V{\cdot}\AA^{-1}}$ (right). C_2mim is coloured green and BF_4 is coloured red, Sourced from Zhao et al..[16]

- proof of concept for ILs - Xu et al.[11]

- Adding h-bonding - [$c_2$mim][OAc] in equilibrium with carbene-acetate form - Brehm et al.[15]

re-read

–H-bonding network in ILs - Canongia Lopes and Pádua[18]

– Ethyl chain to reduce the amount of nonpolar aggregation that would disrupt the H-bonding network - Canongia Lopes and Pádua[18]

– Notion of pK_a is broken in pure IL solutions, as it's an aqueous property, proton transfer happens "orders of magnitude" less - MacFarlane and Seddon[13]

- Power of IL intramolecular interactions

– Dissolution of cellulose? - Swatloski et al.[19] read this article Mohd et al.[20] and this one

1.4 Computational Chemistry

Just wing this section? you know this bit to death... - QM - Cramer[21]

- DFT - Cramer[21]

– empirical dispersion correction for accurate polarisation - Izgorodina et al.[22]

- MD

– since large scale calcs aren't viable

– pMD to account for dispersion effects - Bedrov et al.[23]

1.5 Research Objectives

2 References

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